

Why the Gram–Cosine Separability Metric Misreads the Normalized Trajectories, and an Affine–Overlap Test for Detecting an Additive Term in the Ansatz

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1 Context and motivation

We test whether the computed diffusive–layer density admits the separable (product) form

$$\phi(r, \theta) = R(r) \Theta(\theta), \quad (1)$$

by analyzing the truncated density matrix $D \in \mathbb{R}^{(m-1) \times n}$ (rows = radial rings, columns = angular rays; the absorbing ring at $r = 1$ is dropped). The current diagnostics are: (i) column-normalized cosine similarity of the angular trajectories $A = D^T$ and radial trajectories $B = D$, summarized by the mean/maximum alignment errors

$$E_{\text{mean}} = \frac{\| |C| - 1 \|_F}{\sqrt{N}}, \quad E_{\text{max}} = \max_{i,j} |C_{ij} - 1|, \quad (2)$$

and (ii) the rank-one energy ratio $\eta = \sigma_1^2 / \sum_k \sigma_k^2$.

Empirically these read as “separable” ($E_{\text{mean}} \sim 10^{-4}$, $\eta \approx 0.9998$), yet when the normalized trajectories are plotted and overlaid they visibly fail to coincide: their *shape* changes with the fixed coordinate. This report explains *why* the metric and the plot disagree, and proposes a complementary test that not only quantifies the deviation in a magnitude-faithful way, but is specifically designed to distinguish a **multiplicative** ansatz (1) from one carrying an **additive** term.

Illustrative measurement. A direct solver run (24×24 grid, $N = 4$, $v = 20$, $w = 10$, $T = 0.5$) gives the following, making the discrepancy explicit:

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Angular   E_mean = 8.08e-04   E_max = 2.24e-03
Radial    E_mean = 3.31e-04   E_max = 8.50e-04
Rank-1 energy ratio eta      = 0.99981   (1 - eta = 1.89e-04)
--- but on the normalized overlay ---
Angular worst |cos| = 0.99776 -> 3.83 deg, unit-L2 gap 0.067, max rel. spread
14.8%
Radial worst |cos| = 0.99915 -> 2.36 deg, unit-L2 gap 0.041, max rel. spread
16.9%
(relative spread is ~3% on the high-magnitude inner rings,
~17% on the low-magnitude outer rings near the absorbing boundary)
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A scalar that reads 10^{-4} thus coexists with trajectories that disagree by $\sim 17\%$ pointwise. The following four mechanisms account for this gap.

2 Why the cosine / rank-one diagnostics fail

2.1 The inner product is magnitude-weighted (tail blindness)

For two trajectories x, y (e.g. radial profiles at two angles) the cosine is

$$\cos(x, y) = \frac{\langle x, y \rangle}{\|x\| \|y\|} = \frac{\sum_i x_i y_i}{\|x\| \|y\|}. \quad (3)$$

The sum is dominated by the samples where $x_i y_i$ is large, i.e. the high-density inner rings near the central source. The non-separable structure, however, lives in the low-density outer rings near the absorbing boundary. Write $y_i = R_i(1 + \varepsilon_i)$ with a *relative* mismatch ε_i concentrated where R_i is small; then

$$\langle x, y \rangle - \langle x, x \rangle = \sum_i R_i^2 \varepsilon_i, \quad (4)$$

and the contribution of the tail samples is suppressed by the factor $R_i^2 \rightarrow 0$. A large relative deviation in the tail moves the cosine essentially not at all. The eye, by contrast, reads the normalized curves on a common vertical scale and perceives the *relative* disagreement across the whole domain, tail included. The metric and the plot are answering different questions.

2.2 Two layers of square-root / squared compression

Even where a deviation does register, it is compressed twice over.

Cosine to geometric gap. For unit-normalized trajectories $\hat{x}, \hat{y}$,

$$\|\hat{x} - \hat{y}\|_2^2 = 2(1 - \cos(x, y)) \implies \|\hat{x} - \hat{y}\|_2 = \sqrt{2(1 - \cos)}. \quad (5)$$

Thus $\cos = 0.99915$ corresponds to a unit-vector gap of 0.041 and an angle of only 2.36° : a genuine separation is reported as a near-zero “error.”

Energy versus amplitude in η . The rank-one ratio uses *squared* singular values, so it reports a residual *energy* fraction

$$1 - \eta = \frac{\sum_{k \geq 2} \sigma_k^2}{\sum_k \sigma_k^2}, \quad (6)$$

whereas the relative *amplitude* of the non-separable part is

$$\frac{\|D - \sigma_1 u_1 v_1^T\|_F}{\|D\|_F} = \sqrt{1 - \eta}. \quad (7)$$

For $\eta = 0.99981$ this is $\sqrt{1.89 \times 10^{-4}} \approx 0.014$, i.e. 1.4% in amplitude — two orders of magnitude larger than “ $1 - \eta$ ” suggests, and larger still ($\sim 17\%$) when measured pointwise relative to the local value.

2.3 Diagonal dilution and the absolute value

In (2) the normalization $N = C.\text{size} = k^2$ for a $k \times k$ similarity matrix includes the k diagonal entries, which are *exactly* 1, so $|C_{ii}| - 1 = 0$ contributes k hard zeros to the Frobenius mean and biases E_{mean} downward by the factor $\sqrt{1 - 1/k}$ (and, more importantly, dilutes the off-diagonal signal that actually carries the information). Taking $|C|$ also discards sign, hiding any anti-alignment. These are secondary to §2.1–§2.2 but compound them.

2.4 The deeper issue: the wrong hypothesis class

Column normalization $x \mapsto x/\|x\|$ is a *homogeneous* (pure-scaling) map. Consequently the cosine similarity can only ever test the *multiplicative / proportional* hypothesis

$$c_j = \alpha_j R, \quad (8)$$

i.e. that every trajectory is a scalar multiple of one reference profile. It is *structurally incapable* of representing, and therefore of detecting, an additive offset. If the true field carries a term that is constant in r but varies with θ (or vice versa), normalization partially absorbs it and the cosine remains near unity. A high cosine therefore tells us nothing about whether the ansatz needs an additive component. To ask that question we must enlarge the transform group from *scalings* to *affine maps* (scaling *plus* translation). The extra degree of freedom — the translation — is exactly what senses additivity, and is the subject of the next section.

3 Proposed approach: an affine–overlap test for additivity

3.1 The two competing ansatz families

We wish to discriminate, from the numerics alone, between

$$(M) \text{ multiplicative: } \phi(r, \theta) = R(r) \Theta(\theta), \quad D = R \Theta^T \quad (\text{rank } 1), \quad (9)$$

$$(A) \text{ additive: } \phi(r, \theta) = g(r) + h(\theta), \quad D = g \mathbf{1}_n^T + \mathbf{1}_m h^T \quad (\text{rank } \leq 2), \quad (10)$$

$$(M+A): \quad \phi(r, \theta) = R(r) \Theta(\theta) + g(r) + h(\theta), \quad D = R \Theta^T + g \mathbf{1}_n^T + \mathbf{1}_m h^T. \quad (11)$$

The decisive observation is how an additive term acts *column by column*. Fixing θ_j (one radial trajectory $c_j = D_{:,j}$):

$$\underbrace{c_j = \Theta_j R}_{(M): \text{ scaling only}}, \quad \underbrace{c_j = \Theta_j R + h_j \mathbf{1}_m + g}_{(M+A)}. \quad (12)$$

An additive term $h(\theta)$ that is constant in r appears as a *column-dependent vertical offset* $h_j \mathbf{1}_m$ on top of a scaled copy of the common shape R . This is precisely a per-column **affine** relationship, and it is invisible to the scaling-only cosine test.

3.2 Per-trajectory affine fit

Fix a reference radial shape u (the leading left singular vector u_1 of D , or the highest-norm column). For each radial trajectory c_j solve the two-parameter least-squares problem

$$(\alpha_j, \beta_j) = \arg \min_{\alpha, \beta} \|c_j - \alpha u - \beta \mathbf{1}_m\|_2^2, \quad (13)$$

with closed-form normal equations

$$\begin{bmatrix} \langle u, u \rangle & \langle u, \mathbf{1} \rangle \\ \langle u, \mathbf{1} \rangle & m \end{bmatrix} \begin{bmatrix} \alpha_j \\ \beta_j \end{bmatrix} = \begin{bmatrix} \langle u, c_j \rangle \\ \langle \mathbf{1}, c_j \rangle \end{bmatrix}. \quad (14)$$

We deliberately **do not force the trajectories to coincide**; we *fit* the affine map and then ask whether the offset β_j was *needed*. Compare two nested residuals,

$$S_j^{\text{mult}} = \min_{\alpha} \|c_j - \alpha u\|^2 = \|c_j\|^2 - \frac{\langle u, c_j \rangle^2}{\|u\|^2}, \quad S_j^{\text{aff}} = \|c_j - \alpha_j u - \beta_j \mathbf{1}\|^2, \quad (15)$$

and report the *offset improvement*

$$\Delta_j = \frac{S_j^{\text{mult}} - S_j^{\text{aff}}}{S_j^{\text{mult}}} \in [0, 1]. \quad (16)$$

The symmetric test is performed on the rows (angular trajectories $\rho_i = D_{i,:}$) against a reference angular shape $w = v_1$, yielding offsets b_i and improvements analogous to (16); a needed b_i signals an additive $g(r)$.

Decision rule.

$\beta_j \approx 0, \Delta_j \approx 0$	offset not needed	multiplicative (M)
$\beta_j \neq 0, \Delta_j$ large, α_j small/constant	offset dominates	additive (A)
$\beta_j \neq 0$ and α_j both structured	both needed	product + additive (M+A)

Crucially, one inspects whether β_j (resp. b_i) varies *smoothly and systematically* with θ_j (resp. r_i): a structured offset is a recovered estimate of $h(\theta)$ (resp. $g(r)$), not fitting noise.

3.3 Two equivalent viewpoints (sanity checks)

(i) **Affine cosine is Pearson correlation.** Minimizing over the offset β in (13) is identical to *mean-centering* both c_j and u before measuring proportionality. Hence the scale-free scalar summary of the affine fit is the Pearson correlation

$$\rho(c_j, u) = \frac{\langle c_j - \bar{c}_j \mathbf{1}, u - \bar{u} \mathbf{1} \rangle}{\|c_j - \bar{c}_j \mathbf{1}\| \|u - \bar{u} \mathbf{1}\|}, \quad (17)$$

i.e. the *centered* cosine. The diagnostic for additivity is the *gap* between the raw (uncentered) cosine and the Pearson correlation: if centering markedly improves the alignment, an additive offset is present. This is a one-line addition to the existing pipeline.

(ii) **Global two-way (ANOVA / AMMI) decomposition.** The per-trajectory picture is the local face of the global model

$$D_{ij} = \underbrace{c + g_i + h_j}_{\text{additive part}} + \underbrace{R_i \Theta_j}_{\text{multiplicative interaction}} + \varepsilon_{ij}. \quad (18)$$

Fit the additive part by the classical two-way means $\hat{c} + \hat{g}_i + \hat{h}_j = \bar{D}_{i.} + \bar{D}_{.j} - \bar{D}$, subtract it, and take the SVD of the residual *interaction* matrix E_{ij} . Then:

- additive part negligible, E rank-one and dominant $\Rightarrow$ pure multiplicative (M);
- additive part dominant, E negligible $\Rightarrow$ pure additive (A);
- both sizeable, E rank-one $\Rightarrow$ product + additive (M+A), with R, Θ read off from the leading singular triple of E .

This is exactly Tukey’s “one degree of freedom for non-additivity” / the AMMI model, and it gives a clean energy budget: the fraction of $\|D\|_F^2$ carried by the additive part versus the rank-one interaction directly answers “is there an additive term?”

3.4 Identifiability and pitfalls

- **Reference must not be constant.** If $u \propto \mathbf{1}$ the columns $\{u, \mathbf{1}\}$ are collinear and β_j is unidentifiable. Check $\langle u, \mathbf{1} \rangle^2 < \|u\|^2 m$ (strict) before fitting; near the fully relaxed J_0 regime u flattens and this guard matters.
- **Gauge freedom.** In (18) a constant can move freely between c , g , and h . Fix the gauge (e.g. $\sum_i g_i = \sum_j h_j = 0$) so the recovered g, h are comparable across runs.
- **Do not normalize first.** The additivity signal lives in the offset; column-normalizing (as the cosine pipeline does) destroys it. The affine fit must run on the *raw* D .
- **Report amplitude, not energy.** Quote $\sqrt{1 - \eta}$ and the max pointwise relative residual alongside any η or E_{mean} , so the scalar tracks the plot.

3.5 Connection to the analytic expansion

The eigenfunction expansion already derived,

$$\phi(r, \theta, t) = \sum_{n \in \mathbb{Z}} \sum_j C_{n,j} e^{inN\theta} J_{|n|N}(\kappa_{|n|N,j} r) e^{-\kappa_{|n|N,j}^2 t}, \quad (19)$$

makes the additive-versus-multiplicative question physical. Group the sum as

$$\phi(r, \theta, t) = \underbrace{f_0(r, t)}_{n=0 \text{ mode}} + \sum_{n \neq 0} e^{inN\theta} f_n(r, t), \quad (20)$$

where $f_0(r, t) = \sum_j C_{0,j} J_0(\kappa_{0,j} r) e^{-\kappa_{0,j}^2 t}$ is *angularly constant*: it is precisely an additive radial term $g(r)$ sitting beneath the angular ($n \neq 0$) modes. Hence

- if a single $n \neq 0$ product mode dominates the remainder, the field is well described by $\phi \approx g(r) + R(r)\Theta(\theta)$ — the (M+A) class, and the affine offsets β_j recover $g(r)$;
- if the $n = 0$ mode alone survives (the fully relaxed, angularly symmetric limit), the field is genuinely separable *and* θ -independent, and both the cosine test and the affine test agree.

The proposed affine-overlap test therefore does more than fix a metric: it measures the relative weight of the angularly symmetric ($n = 0$) component against the angular product modes, i.e. it reads off which terms of the analytic ansatz the numerics actually support.

4 Recommended reporting

For each solver snapshot, alongside the existing E_{mean} , E_{max} , and η , report:

1. the amplitude residual $\sqrt{1-\eta}$ and the maximum *pointwise relative* residual of the rank-one reconstruction (magnitude-faithful deviation);
2. the per-trajectory affine offsets $\{\beta_j\}$, $\{b_i\}$ and the offset-improvement Δ of (16), for both radial and angular trajectories;
3. the raw-cosine vs. Pearson-correlation gap (the scalar additivity signal);
4. the AMMI energy budget: fraction of $\|D\|_F^2$ in the additive part versus the leading interaction mode.

Together these distinguish (M), (A), and (M+A) and supply the functional-form ansatz directly from the numerical density.